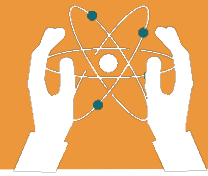


UNIT 7

INTRODUCTION TO KINEMATICS

Learning objectives:

- Understand translatory, rotatory, and vibratory motion.
- Differentiate between distance and displacement with formulas.
- Learn the concepts of speed and velocity, and the difference between scalar and vector quantities.



INTRODUCTION TO KINEMATICS

Kinematics

Kinematics is the study of motion without considering forces.

7.1 Types of Motion

Motion is simply the change in position of an object over time. We will look at three major types of motion:

1. Translatory Motion:

Motion where all parts of an object move in the same direction and distance.

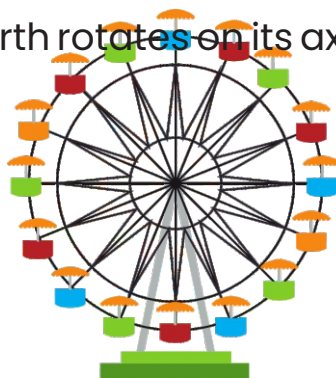
For example: A car moving straight is an example where all parts of the car move forward simultaneously.

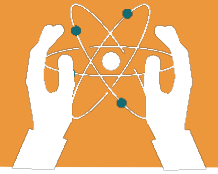


2. Rotatory Motion:

In this type of motion, an object rotates around a fixed point or axis. In this motion the points farthest from the centre move the most.

For example, the Earth rotates on its axis or a wheel spinning.





3. Vibratory Motion:

Vibratory motion is the back-and-forth movement around a fixed position.

For example, A common example is a guitar string vibrating when plucked.



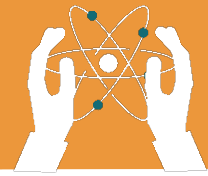
7.2 Scalar and Vector quantities

Scalar Quantities: These are quantities that only have magnitude (size) and no direction.

Examples include distance, speed, time, and mass.

Vector Quantities: These are quantities that have both magnitude and direction.

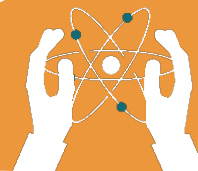
Examples include displacement, velocity, and force.



7.3 Distance vs. Displacement

Distance	Displacement
The total path travelled.	The shortest straight-line distance.
Scalar quantity (only magnitude).	Vector quantity (magnitude + direction).
It is always positive.	It can be zero, negative or positive.
Example: 5m forward + 5m back = 10m.	Example: Displacement = 0m (back to start).
Measured in metres, kilometres, etc.	Measured in metres, kilometres with direction.

INTRODUCTION TO KINEMATICS



Distance (d):

$$\text{Formula: } d = v \times t$$

Where:

d = Distance traveled

v = Speed or velocity

t = Time

Change in Displacement (Δx):

$$\text{Formula: } \Delta x = x_f - x_i$$

Where:

Δx = Displacement

x_f = Final position

x_i = Initial position

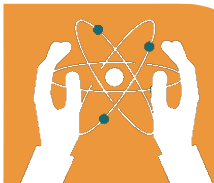
Speed and Velocity:

Speed is the distance travelled per unit of time and is a scalar quantity. Its SI unit is metres per second (m/s).

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Types of Speed

- **Uniform Speed:** Speed remains constant over time.
- **Non-Uniform Speed:** Speed varies over time.



INTRODUCTION TO KINEMATICS

Velocity

Velocity is the rate of change of displacement and is a vector quantity, meaning it has both magnitude and direction.

The formula for velocity is:

$$\text{Velocity} = \frac{\Delta x}{t}$$

Where:

- **v** = velocity
- **Δx** = displacement
- **t** = time

For example, travelling at 10 m/s north is velocity, while 10 m/s is just speed.

Three Types of Velocity

1. Uniform Velocity

Equal displacement in equal time intervals.

2. Non-Uniform Velocity

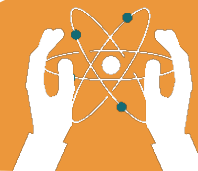
When an object's displacement is unequal in equal intervals of time or its direction changes.

3. Average Velocity

Total displacement divided by total time.

4. Instantaneous Velocity

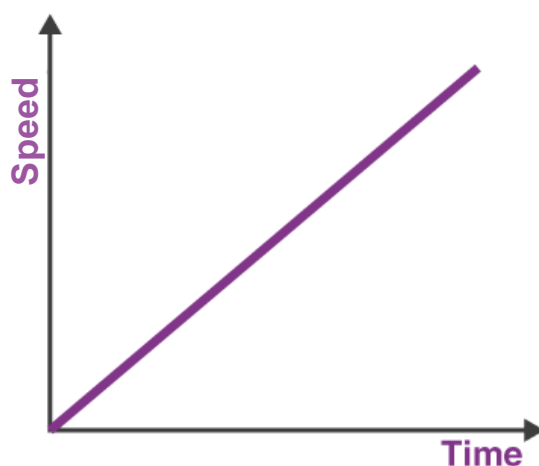
- Velocity at a specific moment.
- Slope of a tangent line on a distance-time graph.



7.4 Graphical Representation of Motion

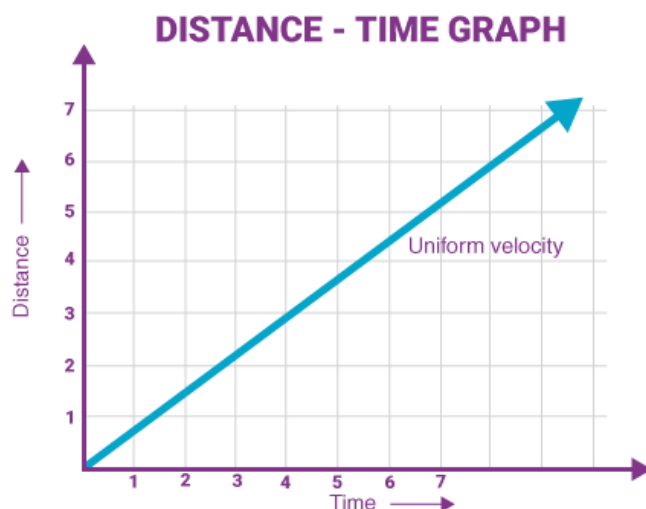
1. Speed-Time Graph

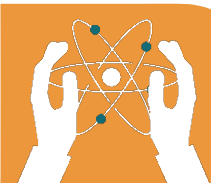
These graphs show how speed changes with time. A horizontal line indicates constant speed, while a sloped line indicates acceleration or deceleration.



2. Distance-Time Graph

These graphs show how distance changes with time. A straight line represents constant speed, while a curve shows changing speed.



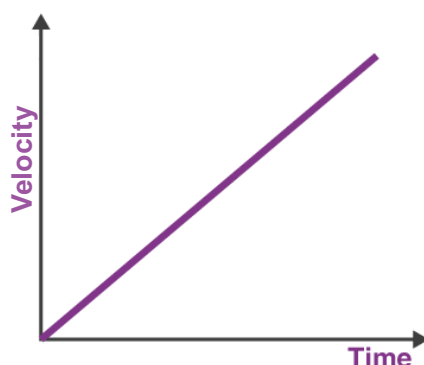


INTRODUCTION TO KINEMATICS

3. Velocity-Time Graph

It shows how velocity changes over time.

- **Horizontal line:** Constant velocity.
- **Sloped line:** Uniform acceleration.
- **Downward sloping line:** Deceleration.



7.5 Problem Solving in Kinematics

In this section, we will solve problems related to motion. First, we will look at simple problems using the formula:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Let's start with a simple problem:

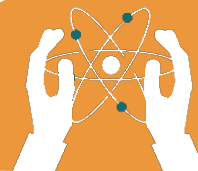
Example: If a car travels a distance of 100 metres in 20 seconds, what is its speed?

Solution:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Speed} = \frac{100}{20}$$

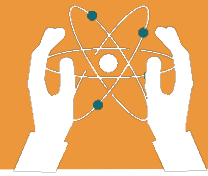
$$\text{Speed} = 05 \text{ m/s.}$$



EXERCISES:

Fill in the blanks.

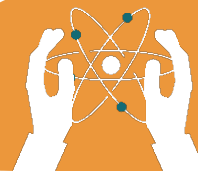
1. The distance travelled by an object is called _____.
2. An object moving in a circular path exhibits _____ motion.
3. The formula for calculating speed is _____ = Distance / Time.
4. _____ is a vector quantity that includes both magnitude and direction.
5. The change in velocity of an object over time is called _____.
6. A moving object that slows down is said to be undergoing _____.
7. The total path length travelled by an object is called _____.
8. In a Distance-Time graph, a straight line indicates _____ speed.



INTRODUCTION TO KINEMATICS

Tick(✓) for True and cross (x) for False statements.

1. Speed is a vector quantity. ☐
2. Displacement is always less than or equal to the distance travelled. ☐
3. A Distance-Time graph with a curved line represents uniform motion. ☐
4. Velocity includes both speed and the direction of motion. ☐
5. In a Speed-Time graph, a horizontal line indicates constant speed. ☐
6. An object in vibratory motion moves back and forth around a fixed position. ☐
7. Distance and displacement are the same when the motion is in a straight line. ☐
8. Uniform acceleration means that the rate of change of velocity is constant. ☐



Choose the correct answer.

1. Which of the following is an example of vibratory motion?

- a) A rotating wheel
- b) A car moving along a straight path
- c) A guitar string vibrating
- d) A satellite orbiting Earth

2. What does a steep slope in a Speed–Time graph represent?

- a) Constant speed
- b) No motion
- c) Acceleration or deceleration
- d) Constant velocity

3. Which of the following is true about displacement?

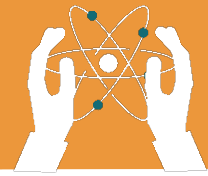
- a) It is always equal to the distance travelled
- b) It depends on the direction of motion
- c) It is a scalar quantity
- d) It is never zero

4. Which of the following describes the type of motion where all parts of an object move the same distance?

- a) Rotatory motion
- b) Vibratory motion
- c) Translatory motion
- d) Circular motion

5. What is the unit of speed?

- a) Metre
- b) Metre/second
- c) Second
- d) Kilometre



INTRODUCTION TO KINEMATICS

6. The formula $V = \Delta x/t$ used to calculate _____.

- a) Displacement b) Speed
- c) Acceleration d) Velocity

7. What happens to the speed of an object when it accelerates?

- a) The speed decreases b) The speed remains constant
- c) The speed increases d) The speed becomes zero

8. If an object covers 50 metres in 10 seconds, its speed is _____.

- a) 5 m/s b) 50 m/s
- c) 0.5 m/s d) 500 m/s

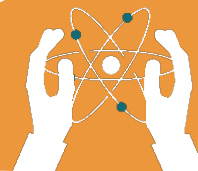
Answer the Following Questions:

Short-Answer Questions.

1. What is kinematics?
2. Write the differences between scalar and vector quantities.
3. Write the differences between distance and displacement.
4. What is speed? Write its formula and unit.
5. What is velocity? Write its formula and unit.
6. Define distance and displacement with an example.

Long-Answer Questions.

1. What is motion? Describe the types of motion with examples.



Create a Simple Distance–Time Graph



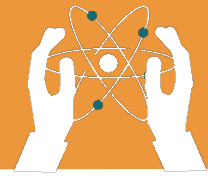
ACTIVITY

- **Materials:** Toy car, measuring tape, stopwatch, and graph paper.
- **Instructions:** Students will use a toy car to measure distance travelled over time and plot the results on a distance–time graph. This project helps visualize constant speed and interpret motion through graphs.



Critical Thinking

Imagine you are driving a car. How would the Distance–Time graph look if you were driving at a constant speed, and how would it change if you suddenly stopped?



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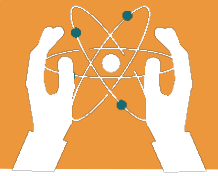
Numericals:

- A car travels at a constant speed of 15 metres per second for 8 seconds. How far does the car travel?
- A cyclist covers a distance of 150 metres in 12 seconds. What is the cyclist's velocity?
- A person walks 10 metres north, then turns around and walks 4 metres south. What is their displacement?

Q1. Word problems

- a)** A zombie is ambling down the road and travels 17 km over the course of a 12-hour period. What is Zombie Zed's average speed in m/s?
- b)** An object is moving with an initial velocity of 5 m/s. It accelerates at a rate of 2 m/s^2 for 8 seconds. What is its final velocity?

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- Match the motion descriptions 1–4 below to the correct motion diagrams A–D by filling in the appropriate letter.
- State the reason for your answer in the third column.

A.



B.

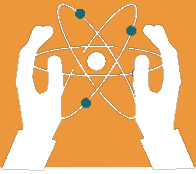


C.



D. None of the above

Description	Diagram	Reason
1. At Constant Speed	_____	_____
2. At Rest	_____	_____
3. Slowing Down	_____	_____
4. Speeding Up	_____	_____

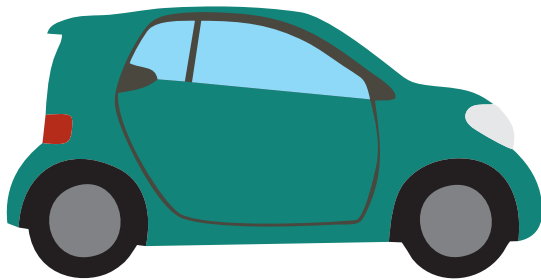


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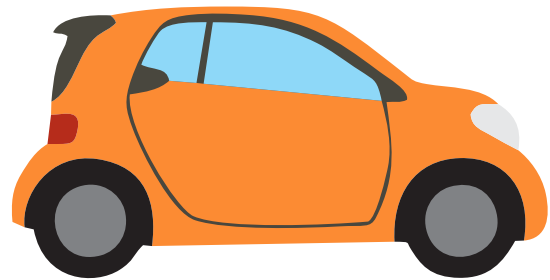
Acceleration

$$a_{\text{Acceleration}} = \frac{\Delta v_{\text{Change in Velocity}}}{\Delta t_{\text{Change in Time taken}}}$$

This Car is
Slowing Down

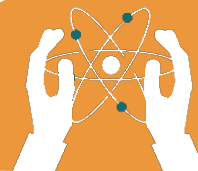


This Car is
Speeding Up



Negative acceleration is
also called deceleration.

INTRODUCTION TO KINEMATICS



Fill in the blanks.

No	Initial Velocity	Final Velocity	Time	Acceleration
1.	0 m/s	60 m/s	10s	
2.	10 m/s	55 m/s	15s	
3.	0 m/s	40 m/s	4s	
4.	60 m/s	40 m/s	10s	
5.	20 m/s	6 m/s	2s	

6. A downhill skier picks up speed from 0 m/s to 40 m/s in 4 seconds. What is the acceleration? _____

7. A cruise ship coasts down from 30 m/s to 0 m/s in 8 minutes. What is the acceleration? _____

8. A train speeds up from 5 m/s to 20 m/s in 50 seconds. What is the acceleration? _____

9. A skateboarder speeds up to 20 m/s in 4 seconds. What was the acceleration? _____

10. A car rolls backward from -2 to -10 m/s in 4 seconds. What is the acceleration? _____